ZOMATO RESTAURANT ANALYTICS

<u>SQL + Power BI End-to-End Business Intelligence Project</u>

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Tools Used:
• MySQL
• Power BI
• Excel
• GitHub

Dataset : Zomato Restaurants Dataset (9551 Restaurants):
https://www.kaggle.com/datasets/shrutimehta/zomato-restaurants-data

SECTION 1: PROJECT OVERVIEW

Project Overview

The objective of this project was to analyze restaurant data from Zomato using SQL and Power BI. The dataset contains information about restaurants across multiple cities, including ratings, cuisines, pricing, online delivery availability, table booking services, and customer votes.

The project was divided into two major phases:

1. SQL Analysis for extracting business insights from raw data.
2. Power BI Dashboard Development for visual storytelling and decision-making.

The final outcome is an interactive analytics dashboard that enables business users to understand customer preferences, city-wise restaurant distribution, cuisine popularity, pricing trends, and delivery service performance.

Section 2: Dataset Description

Attribute	Description
Restaurant ID	Unique identifier
Restaurant Name	Restaurant name
City	Restaurant location
Aggregate Rating	Customer rating
Average Cost for Two	Pricing indicator
Cuisines	Cuisine category
Has Online Delivery	Delivery availability
Has Table Booking	Reservation availability
Votes	Customer votes

Section 3: SQL Analysis

Phase 1: Data Cleaning & Validation

Business Question:-

Q) How complete and reliable is the dataset before analysis?

```
SELECT

COUNT(*) AS total_rows,

SUM(city IS NULL) AS missing_city,

SUM(cuisines IS NULL) AS missing_cuisines,

SUM(aggregate_rating IS NULL) AS missing_rating

FROM zomato_clean;
```

Key Insight

Identify whether missing values could bias later conclusions.

Phase 2: Exploratory Data Analysis (EDA)

Query 1: Restaurants by City

Business Question: Q) Which cities have the highest restaurant concentration?

```sql
-- RESTAURANTS PER CITY --
SELECT city,
COUNT(*) AS restaurant_count
FROM zomato_clean
GROUP BY city
ORDER BY restaurant_count DESC;
```

Query 2: Average Rating by City

Business Question: Q) Which cities have the highest customer satisfaction levels?

```sql
-- AVERAGE RATING PER CITY --
SELECT city,
ROUND(AVG(aggregate_rating),2) AS avg_rating
FROM zomato_clean
WHERE aggregate_rating > 0
GROUP BY city
ORDER BY avg_rating DESC;
```

Query 3: Total Customer Engagement

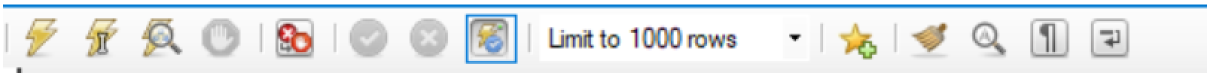
Business Question: Q) Which cities generate the highest customer engagement?

```sql
-- TOP CITIES BY VOTES --
SELECT city,
SUM(votes) AS total_votes
FROM zomato_clean
WHERE votes > 0
GROUP BY city
ORDER BY total_votes DESC;
```

Phase 3: Core Business Analytics

Query 4: Top-Rated Cuisines

Business Question: Q) Which cuisines consistently receive the highest customer ratings?




```sql
-- TOP RATED CUISINES --

SELECT cuisines,
COUNT(*) AS restaurants,
ROUND(AVG(aggregate_rating),2) AS avg_rating
FROM zomato_clean
WHERE aggregate_rating > 0
GROUP BY cuisines
HAVING COUNT(*) >= 10 -- removes cuisines with very few restaurants --
ORDER BY avg_rating DESC;
```

Query 5: Online Delivery Impact

Business Question: Q) Does offering online delivery improve customer engagement and ratings?

```sql
-- ONLINE DELIVERY IMPACT --

SELECT has_online_delivery,
COUNT(*) AS restaurants,
ROUND(AVG(aggregate_rating),2) AS avg_rating,
ROUND(AVG(votes),0) AS avg_votes
FROM zomato_clean
GROUP BY has_online_delivery;
```

Query 6: Restaurant Type Performance

Business Question: Q) Which restaurant categories attract the highest customer engagement?

```sql
-- PRICE RANGE ANALYSIS --
-- 4: LUXURY
-- 3: PREMIUM
-- 2: MODERATE
-- 1: BUDGET

SELECT price_range,
COUNT(*) AS restaurants,
ROUND(AVG(aggregate_rating),2) AS avg_rating,
ROUND(AVG(votes),0) AS avg_votes
FROM zomato_clean
GROUP BY price_range
ORDER BY avg_votes DESC;
```

Phase 4: Window Functions & Ranking

Query 7: Restaurant Ranking within City

Business Question: Q)Within each city, which restaurants rank highest by customer ratings?

```
-- RESTAURANT RANKING WITHIN EACH CITY --
SELECT
restaurant_name,
city,
aggregate_rating,
votes,
RANK() OVER(
PARTITION BY city
ORDER BY aggregate_rating DESC
) AS city_rank
FROM zomato_clean;
```

Query 8: Compare Ranking Functions

Limit to 1000 rows

```
SELECT
restaurant_name,
city,
aggregate_rating,

RANK() OVER(
PARTITION BY city
ORDER BY aggregate_rating DESC
) AS rank_value,

DENSE_RANK() OVER(
PARTITION BY city
ORDER BY aggregate_rating DESC
) AS dense_rank_value,

ROW_NUMBER() OVER(
PARTITION BY city
ORDER BY aggregate_rating DESC
) AS row_num

FROM zomato_clean;
```

PHASE 5: ADVANCE BUSINESS SEGMENTATION

Query 9: Best Value Restaurants

```sql
-- BEST VALUE RESTAURANTS --
-- CTE--
WITH value_restaurants AS (
SELECT
restaurant_name,
city,
aggregate_rating,
average_cost_for_two,
votes
FROM zomato_clean
WHERE aggregate_rating >= 4
AND average_cost_for_two <= 500
)

SELECT *
FROM value_restaurants
ORDER BY aggregate_rating DESC;
```

Query 10: Customer Segmentation

```sql
-- PREMIUM VS BUDGET RESTAURANTS --

SELECT restaurant_name,
city,
average_cost_for_two,
CASE
    WHEN average_cost_for_two >= 2000 THEN 'Premium'
    WHEN average_cost_for_two >= 1000 THEN 'Mid Range'
    ELSE 'Budget'
    END AS Segment

FROM zomato_clean;
```

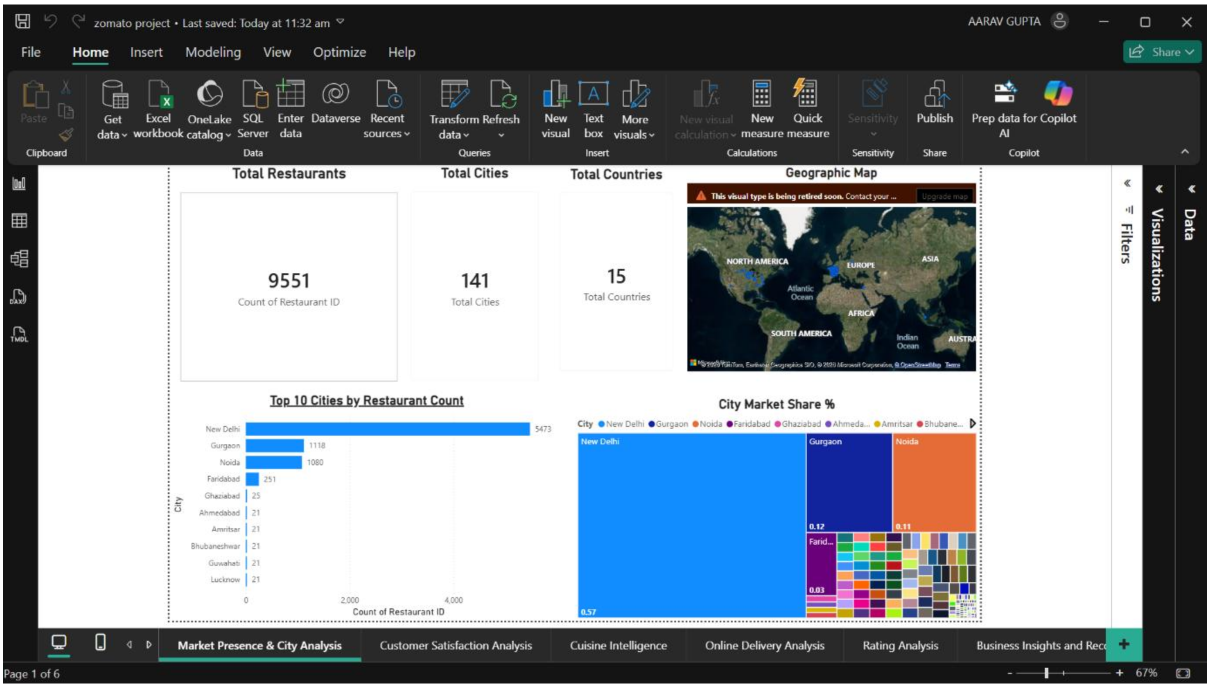
Section 4: SQL Insights Summary

Key SQL Findings

1. New Delhi contains the highest number of restaurants.
2. Online delivery services are available in only a subset of restaurants.
3. Highly rated restaurants are not always the most expensive.
4. Customer votes are positively correlated with restaurant ratings.
5. Certain cuisines dominate the market share.

Section 5: Power BI Dashboard

Dashboard 1 – Market Presence and City Analysis



Objective

To understand the overall restaurant landscape, market concentration, and geographical distribution of restaurants across different cities.

Business Questions Answered

- How many restaurants are listed on Zomato?
- Which cities have the highest number of restaurants?
- How many cities are covered in the dataset?

- Which markets dominate the restaurant industry?

Visualizations Used

- KPI Card – Total Restaurants
- KPI Card – Total Cities
- Bar Chart – Top 10 Cities by Restaurant Count
- Treemap – City Market Share

Key Insights

- New Delhi has the largest restaurant presence with 5,473 restaurants.
- Gurgaon and Noida follow as secondary restaurant hubs.
- Restaurant distribution is highly concentrated in a few major cities.
- Several cities have fewer than 50 listed restaurants.

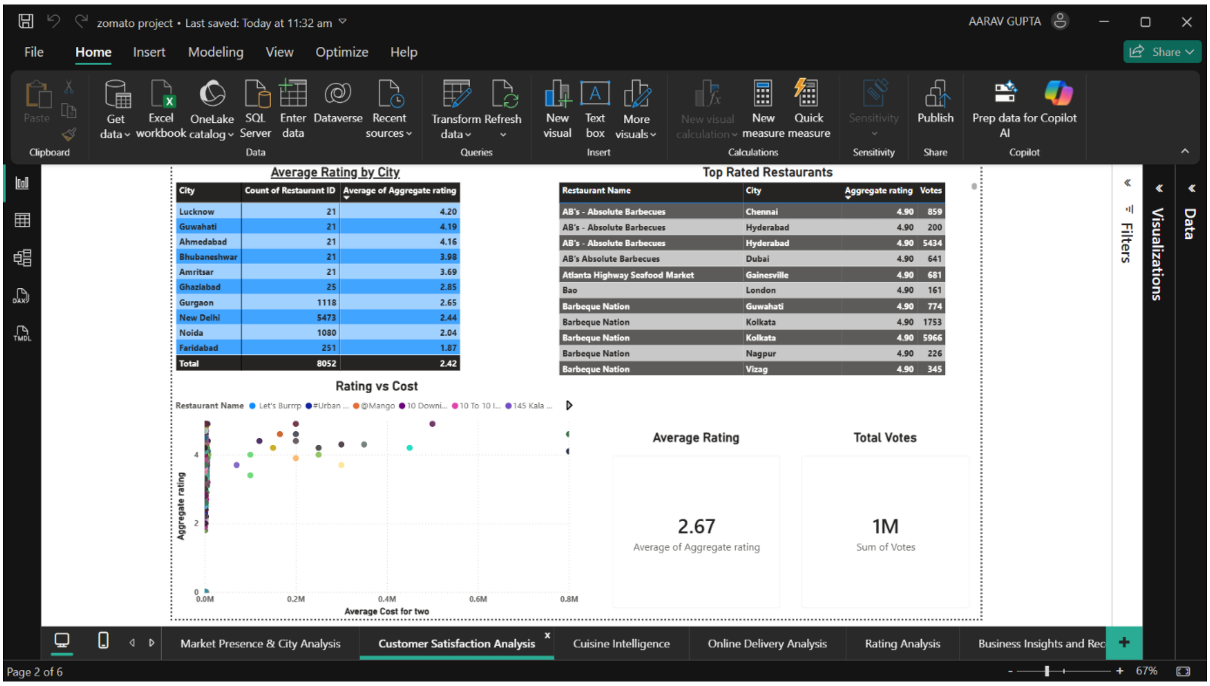
Business Recommendations

- Expand restaurant acquisition efforts in underrepresented cities.
- Focus marketing campaigns on high-density markets.
- Develop localized strategies for smaller cities.

Skills Demonstrated

- Aggregation Analysis
- Market Segmentation
- KPI Development
- Business Intelligence Reporting

Dashboard 2 : Customer Satisfaction Analysis



Objective

To analyze customer satisfaction patterns and identify high-performing markets based on ratings and customer feedback.

Business Questions Answered

- Which cities have the highest average ratings?
- What is the distribution of restaurant ratings?
- Which locations deliver the best customer experience?
- Are highly rated restaurants concentrated in specific cities?

Visualizations Used

- Bar Chart – Top 10 Cities by Average Rating
- Column Chart – Rating Distribution
- KPI Card – Overall Average Rating
- Table – City-wise Rating Performance

Key Insights

- Several smaller cities outperform major metropolitan areas in customer ratings.

- Most restaurants fall within the "Good" and "Very Good" categories.
- Only a small percentage achieve "Excellent" ratings.
- Customer satisfaction varies significantly across locations.

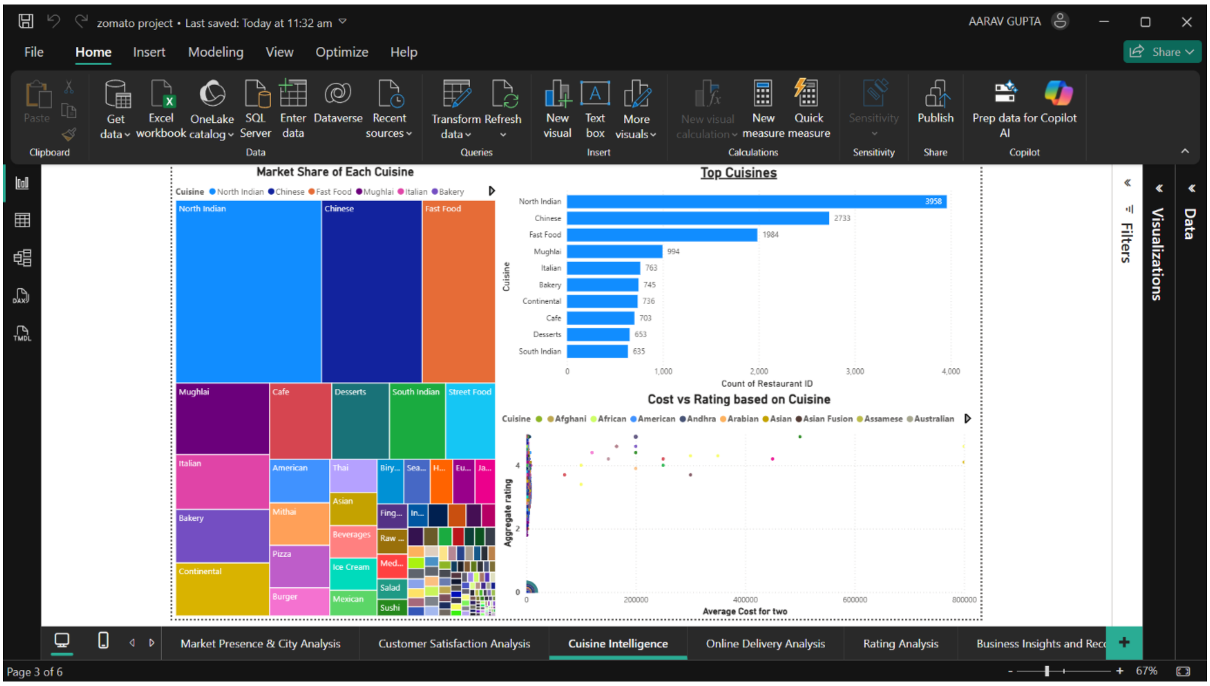
Business Recommendations

- Replicate practices of high-performing cities.
- Identify factors driving excellent ratings.
- Improve service quality in low-performing markets.

Skills Demonstrated

- Rating Analysis
- Customer Experience Analytics
- Comparative Performance Evaluation
- Data Interpretation

Dashboard 3: Cuisine Analysis



Objective

To identify the most popular cuisines, evaluate cuisine performance, and understand customer preferences.

Business Questions Answered

- Which cuisines are most popular?
- Which cuisines receive the highest ratings?
- Which cuisine categories dominate the market?
- What food preferences exist across customers?

Visualizations Used

- Bar Chart – Top 10 Cuisines by Restaurant Count
- Bar Chart – Top Rated Cuisines
- Treemap – Cuisine Market Share
- Scatter Plot – Cuisine Cost vs Rating

Key Insights

- A few cuisines dominate restaurant listings.
- Popular cuisines are not always the highest rated.
- Certain niche cuisines receive exceptionally high ratings.
- Customer demand is concentrated around a limited number of cuisine categories.

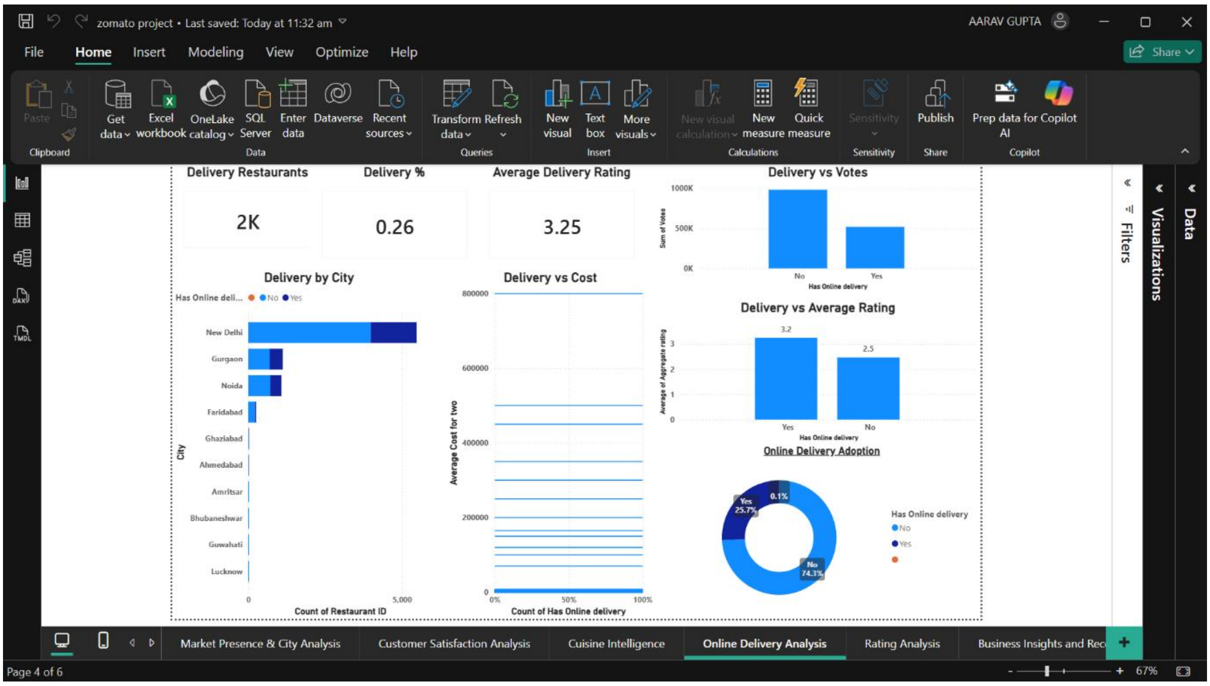
Business Recommendations

- Increase promotion of highly rated niche cuisines.
- Expand successful cuisine categories into new markets.
- Focus on cuisines with high ratings and strong demand.

Skills Demonstrated

- Customer Preference Analysis
- Market Share Analysis
- Category Performance Evaluation
- Strategic Business Insights

Dashboard 4: Delivery & Service Analysis



Objective

To evaluate how online delivery and table booking services impact restaurant performance and customer satisfaction.

Business Questions Answered

- Do restaurants offering online delivery achieve better ratings?
- Does table booking improve customer experience?
- What percentage of restaurants offer delivery services?
- What service features contribute to higher ratings?

Visualizations Used

- Bar Chart – Delivery vs Average Rating
- Bar Chart – Table Booking vs Average Rating
- Donut Chart – Delivery Availability
- Donut Chart – Table Booking Availability
- KPI Cards – Delivery % and Table Booking %

Key Insights

- Restaurants with online delivery generally receive higher ratings.

- Table booking services are associated with improved customer satisfaction.
- A significant portion of restaurants still do not offer delivery services.
- Convenience factors strongly influence customer perception.

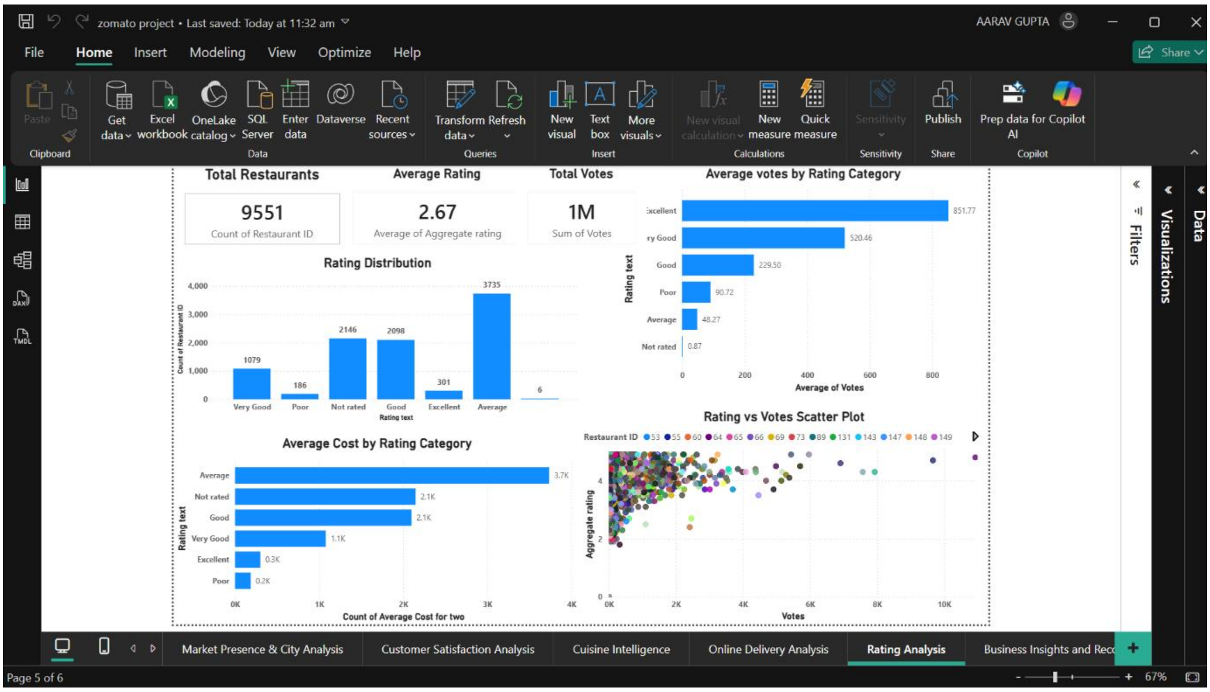
Business Recommendations

- Encourage restaurants to adopt delivery services.
- Promote reservation systems.
- Improve customer convenience features.

Skills Demonstrated

- Service Performance Analytics
- Customer Behavior Analysis
- Operational Insights
- KPI Tracking

DASHBOARD 5: RATING ANALYSIS



Objective

To analyze customer satisfaction levels across restaurants and identify factors associated with higher ratings.

Business Questions Answered

- What is the overall average restaurant rating?
- How are restaurants distributed across different rating categories?
- Is there a relationship between customer votes and ratings?

Visualizations Used

- KPI Cards
- Clustered Column Chart
- Horizontal Bar Chart
- Scatter Plot

Key Insights

- Most restaurants fall within the "Good" and "Very Good" rating categories.
- This indicates a generally positive customer experience.
- Only a small percentage of restaurants achieve "Excellent" ratings.
- These restaurants represent benchmark performers.
- Certain cities consistently outperform others in average ratings.
- These cities can be studied to identify best practices.

Business Recommendations

- Promote highly-rated restaurants through featured listings.
- Investigate service practices of top-performing restaurants and cities
- Encourage low-rated restaurants to improve service quality.

Skills Demonstrated

- Aggregate Functions
- GROUP BY Analysis
- Customer Satisfaction Analytics
- KPI Design
- Comparative Analysis
- Scatter Plot Interpretation
- Business Intelligence Reporting

Business Insights & Recommendations

Objective

To consolidate analytical findings and provide strategic recommendations for restaurant growth and customer satisfaction improvement.

Business Questions Answered

- What are the major findings from the dataset?
- Which business areas require improvement?
- What strategies can improve restaurant performance?
- How can customer satisfaction be increased?

Visualizations Used

- KPI Cards
- Rating Distribution Analysis
- Cost vs Rating Scatter Plot
- City Performance Matrix
- Recommendation Panel
- Executive Summary Text Box

Key Insights

- New Delhi dominates the restaurant market.
- Online delivery positively influences ratings.
- Moderate pricing often achieves ratings comparable to premium restaurants.
- Customer experience has a greater impact than price.
- High-performing cities can serve as benchmarks.

Business Recommendations

- Expand delivery services in key markets.
- Improve customer experience initiatives.
- Promote top-performing cuisine categories.
- Increase adoption of table booking systems.
- Replicate successful city-level strategies.

Skills Demonstrated

- Business Intelligence
- Executive Reporting
- Strategic Decision Support
- Data Storytelling

CONCLUSION

This project demonstrates the application of SQL and Power BI to solve real-world business problems in the restaurant industry. Through data exploration, customer rating analysis, cuisine evaluation, and service performance assessment, actionable insights were generated to support data-driven decision making.

The project showcases skills in SQL querying, data cleaning, data visualization, dashboard design, business analytics, and strategic recommendation development.